Tracking Body Movements While Undergoing a Sit-to-Stand Maneuver

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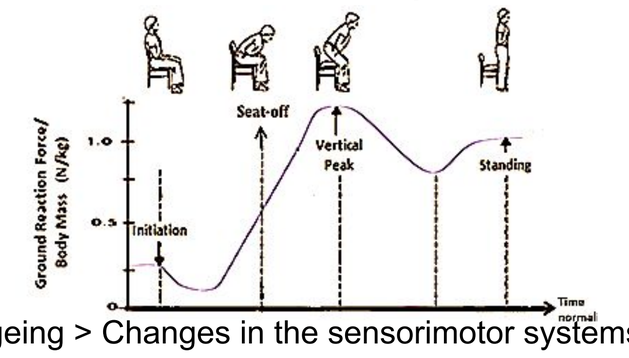
Introduction

StS is the most common daily functional activity.



Ageing > Changes in the sensorimotor systems > Affect on StS maneuver> Can lead to falls

STS can be tracked using a motion capture system, but the clinical deployment of biomechanical motion assessment is limited by the required hardware, time, and expertise.

Goals & Objectives

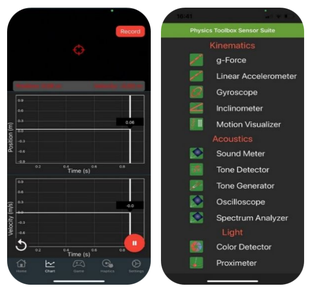
New Method of a Motion Capture System Based on LiDAR sensor

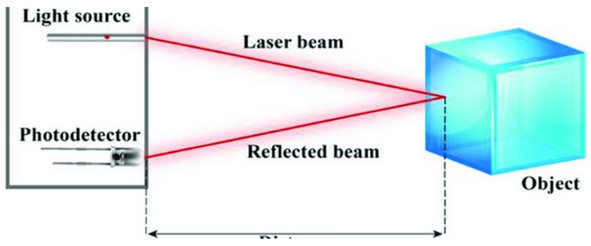
- Portable
- Cheap
- Easy to use

Research evidence for the safety & effectiveness of the 'SitnStand' device

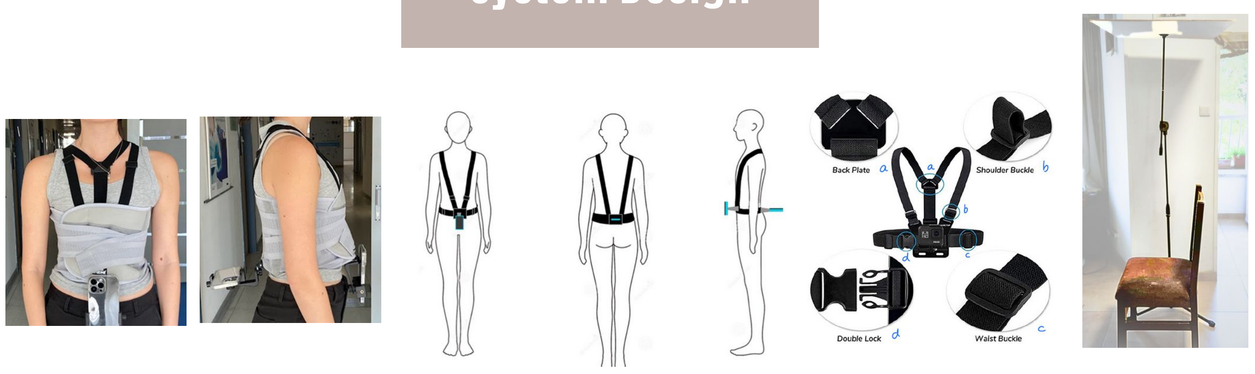
Methods

LiDAR sensor





System Design

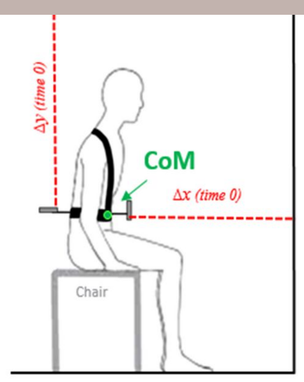


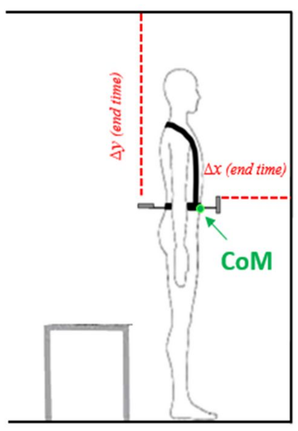
Model Assumptions

$$min\int |\Delta[CoM]X(t)|dt$$

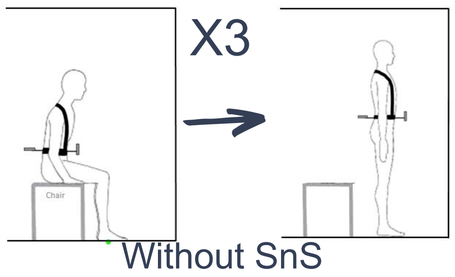
$$min\{\frac{d(\Delta[CoM]X(t))}{dt}\}$$

$$min\{\frac{d^2(\Delta[CoM]X(t))}{dt^2}\}$$



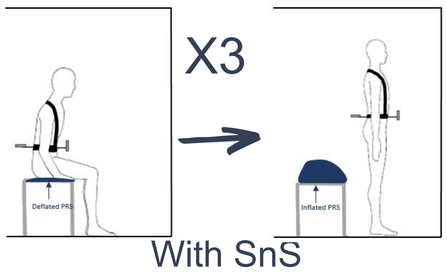


Experiment

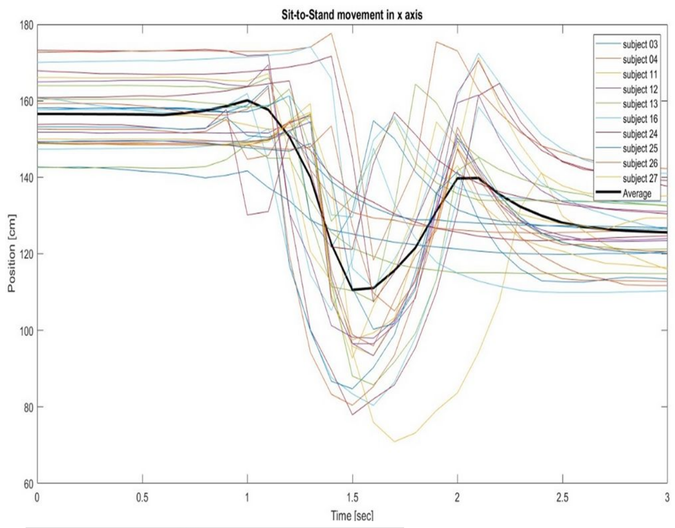


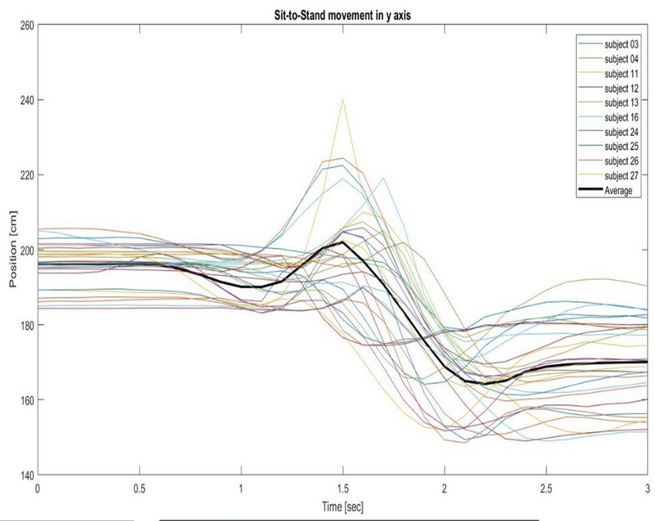
10 Subjects

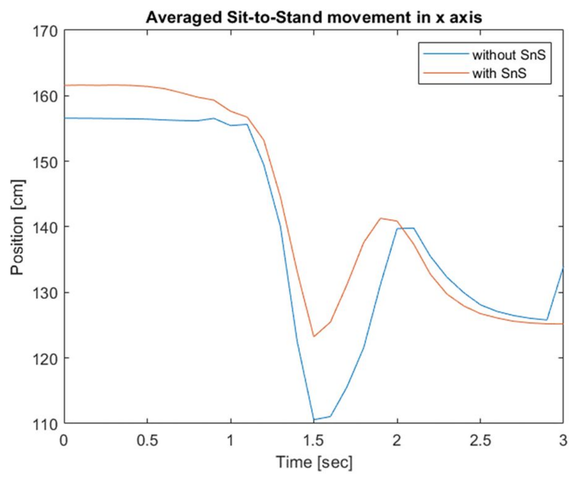
63-72 years old

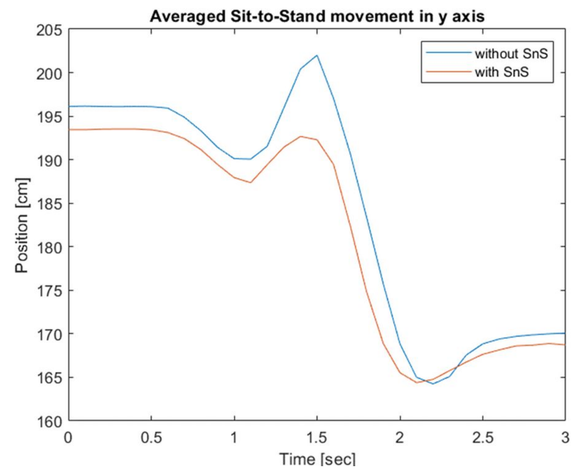


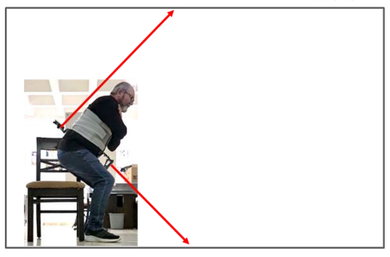
Results Analysis

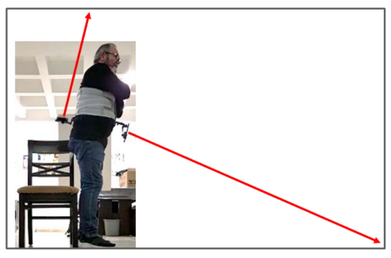


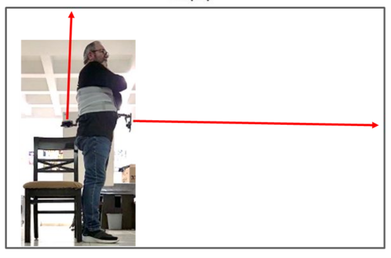












P-Value

Computation Assumption	Axis x [%]	Axis y [%]
AUC	50	70
Velocity	40	30
Acceleration	10	30

Relative Difference

Computation Assumption	Axis x [%]	Axis y [%]
AUC	3.0229	-1.8281
Velocity	3.5866	22.1388
Acceleration	1.2790	28.7940

Conclusions

(1) The LiDAR sensor (in iPhone) can be used as Motion Capture Systems, when the sensor's subject to its limitations.
(2) The computational assumption AUC seems to be the most relevant, especially on the y-axis, to examine stability or instability.
(3) There is no unequivocal statement about the improvement of stability using SnS device.

SitnStand
Portable Smart Rising Seats

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